

Progress Report Sprint 11

- Course Name: CS463 Senior Software Engineering Project III
- Date: 04/10/2026
- Team Name: TaxonBodyMassML
- Team ID: CS.032
- Student Names and Emails:
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Summary - Outcomes, blockers, and insights

- **UI:** Continued to make informed changes to the UI based off of last term's second-phase interviewing and on a heuristic evaluation conducted earlier this sprint
- Started working with project partner to get the hardware we require to host our regressor microservice
- A fully functional version of the confidence interval implementation in both the regressor microservice and the XGBoost training script.

Blockers:

- Our regressor microservice is still not functioning; we have come up with a possible solution, but have not implemented it

Work Log

1. Session History Feature

Status: Complete

Owner: Hailey

Description: Continued implementing suggestions from second-phase interviewing; this time, I implemented a session history feature, with which a user can view all the masses they have generated on one convenient screen.

Source Link: [W 2026-03-12-CS.032-Sprint10-ProgressReport.docx](#)

Supporting Artifact: Item.01 and Item.02

Notes: The output will change when we fix the regressor microservice, but the skeleton is complete.

2. Heuristic Evaluation-Informed UI Updates

Status: Complete

Owner: Hailey

Description: Conducted a heuristic evaluation of the UI based on the 10 usability heuristics. Determined a fix in formatting for the help page as well as the need to introduce a “previous” button onto each tutorial tool tip to allow users to go back. Implemented both of these fixes fully.

Source Link: [☰ CS352 - Heuristic Evaluation](#)

Supporting Artifact: Item.03

3. Confidence Interval for XGBoost Regressor

Status: In-progress

Owner: Grant

Description: This serves as a 90% confidence interval for the XGBoost Regressor. The percentage of the confidence interval can be changed, but this requires rerunning the training script. The API returns a high value, a low value, and a value for confidence. There are many ways that this kind of program can be implemented using XGBoost, but this one only uses a singular version of the model, instead relying on calibration data derived from the results of running the training data through the model. At this point the code is functional, but in a secondary python file for both the regressor microservice and the training script.

Source Link: [Sprint 11 Doc](#)

Supporting Artifacts: Item.04

Notes: The response API has changed, so we will need to update the connection with the UI. However, the request UI remains the same.

Risk and Quality Tracking

Open Bug Count: 4 [☰ TaxonBodyMassML Risk Tracker](#)

TOP BUGS	SEVERITY	OWNER(S)	STATUS	LINK	ACTIONS
NON-RESPONSIVE CSS	Low	Hailey	In-progress	primary CSS file on GitHub	Hailey will add in CSS that better accommodates smaller screen sizes (media queries) by the end of this term (not a top priority)
CACHED QUESTIONS ARE NOT LOADING	Low	Hailey	In-progress	'More questions' service python file on GitHub	Hailey will debug the API call and JS code and then, if necessary, see if it is an

					issue with the render deployment
REGRESSOR MICROSERVICE TOO BIG FOR RENDER FREE TIER	High	Grant and Hailey	In-progress	Render deployment status	We will contact the project partner about accessing dedicated hardware with enough on board memory to handle the Regressor Microservice.
NO ERROR MESSAGES FOR UNMATCHED SPECIES	High	Hailey	In-progress	Source of microservice calls	This can only be completed after the previous bug is patched. After we have both functioning microservices, we can get rid of the taxon display feature and reobtain the error messages.

Next Sprint Plan

- We need to fix our regression microservice deployment and get body mass predictions displayed to the front-end. We have decided on a fix revolving around hardware, though we still need to work with Mark on purchasing the tech. This is a large part of the project expectations from our project partner, so this is something we are focusing on immediately.
Priority: High; Links: [2025-10-23-CS.032-Requirements](#) and [Risk Tracker](#)
- We will begin work on incorporating multi-species predictions into both the lookup table and the regressor microservice. This is in line with REQ-003 which indicates that the user must be able to make at least 10 simultaneous queries.
Priority: Med; Link: [2025-10-23-CS.032-Requirements](#) (REQ-003)

Team Reflection

Overall, we were able to get a decent amount of work done this sprint, considering that it is the first sprint of the term. We were able to fulfill one of the major requirements of our project and made progress in several other places as well.

The biggest priority for the next sprint is to begin working on a solution for the regressor microservice. We have come to the conclusion that a hardware option is likely the best solution, so we will proceed with that in mind. The goal will be to secure dedicated hardware such as a raspberry pi or similar small computer to run the regressor microservice.

Furthermore, we will continue to work on refining the UI according to feedback from our project partner. We have a meeting scheduled to have a focused work session on the LLM based predictor. Lastly, we will continue work on implementing the multi prediction functionality in both the regressor microservice and the lookup table microservice.

Artifacts Appendix

Item.01

[TaxonBodyMass Third-Party UI Interview](#)

This is the document I have used to track questions and recommendations during both phases of interviewing; some things on this document have not been implemented yet, hence why I considered this phase of interviewing “in-progress”.
This sprint, I implemented the “session history” function suggestion.

Item.02

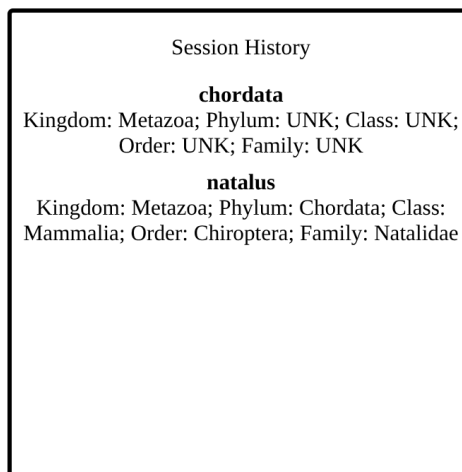


Fig.01 - This is the new session history modal. The user has the option to open this modal and view all results from that session. This information is removed upon refresh and duplicates are not supported.

Item.03

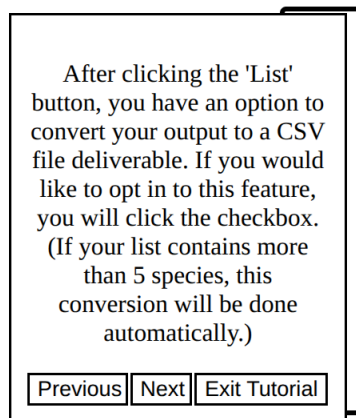


Fig.02- This depicts the new standard for tool tips; each tool tip (barring the very first one) now gives the user the option to go back to the previous tool tip instead of just forward. This adheres to common usability practices.

Item.04

```

Expected mass: 2000.0
Prediction Result:
{
  "confidence": 0.9,
  "lower_bound": 374.6805114746094,
  "prediction": 1693.4993896484375,
  "taxonomy": {
    "class": "Mammalia",
    "family": "Muridae",
    "genus": "Mallomys",
    "kingdom": "Animalia",
    "order": "Rodentia",
    "phylum": "Chordata",
    "species": "Mallomys gunung"
  },
  "upper_bound": 7654.36181640625
}
Expected mass: 40.5580098030442
Prediction Result:
{
  "confidence": 0.9,
  "lower_bound": 10.164824485778809,
  "prediction": 45.94347381591797,
  "taxonomy": {
    "class": "Aves",
    "family": "Cuculidae",
    "genus": "Cacomantis",
    "kingdom": "Animalia",
    "order": "Cuculiformes",
    "phylum": "Chordata",
    "species": "Cacomantis"
  },
  "upper_bound": 207.65756225585938
}
Expected mass: 308.0
Prediction Result:
{
  "confidence": 0.9,
  "lower_bound": 201.4924774169922,
  "prediction": 910.715576171875,
  "taxonomy": {
    "class": "Aves",
    "family": "Megapodiidae",
    "genus": "Megapodius",
    "kingdom": "Animalia",
    "order": "Galliformes",
    "phylum": "Chordata",
    "species": "Megapodius laperouse"
  },
  "upper_bound": 4116.29736328125
}

```

Fig.03 - This is the results of using the test script with the updated regressor microservice. As can be seen, each response has values for confidence, lower bound, upper bound, and prediction value.